

CHAPTER

13



Heart conditions
(like heart failure,
inflammation
or arrhythmia)



Lung conditions
(like asthma, COPD
or infections)



Anemia



Anxiety

Shvasa

Learning Objectives

- ⇒ To learn the concept of Shvasa
- ⇒ To learn the Nidana and Samprapti of Shvasa mentioned in Ayurveda literatures
- ⇒ To frame out the probable Samprapti Ghataka based on the progression of the disease
- ⇒ To learn about the types of Shvasa
- ⇒ To learn about Purvarupa of the disease with current understanding
- ⇒ To learn about specific clinical features of Tamaka Shvasa with its Avastha Bheda (stage of the disease)
- ⇒ To know about the Dosha dominance in type of Shvasa
- ⇒ To know the cardinal feature of each of the type of Shvasa with current understanding
- ⇒ To learn about Sadhyasadyata of the disease with current understanding
- ⇒ To make the differential diagnosis of types of Shvasa

Introduction



Heart conditions
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Anemia



Anxiety

Figure 27: Pratyatma Lakshana of Shvasa

Literally, the term *Shvasa* refers to normal breathing. Another reference says *Shvasa* is a type of 13 *Adharaniya Vega* (a reflex mechanism which should not be suppressed by an individual). In the present context, *Shvasa* (see Figure 27) is a abnormal breathing where the person will be aware of his breathing pattern. This is a disease which

is having the vitiation of *Vata* and *Kapha* Dosha, affliction of *Rasa Dhatu* is seen and involvement of *Pranavaha* and *Rasavaha Srotas* is noted during the pathogenesis of the disease. As *Pranavaha Srotas* affliction is the primary pathogenesis, one can observe the affliction of *Hridaya* and *Mahasrotas* in due course of time because *Srotomula* affliction can be commonest event that can in case of any long standing disease involving any particular *Srotas*.

श्वासत्वं वेगवदूर्ध्ववातत्वं:

Madhu, Ma.Ni. – 12/15-1

“श्वासस्तु भस्त्रिकाध्मानसमवातोर्ध्वगामिता” Madhu, Ma.Ni. – 12/15-1

कफ-वातात्मकावेतौ पित्तस्थानसमुद्भवौ।

हृदयस्य रसादीनां धातूनां चोपशोषणौ॥8॥

Ca.Ci. – 17/8

Nidana of Shvasa

रजसा धूमवाताभ्यां शीतस्थानाम्बुसेवनात्।

व्यायामानाम्यधर्माध्वरुक्षान्नविषमाशनात्॥1॥

आमप्रदोषादानाहाद्रौक्ष्यादत्यतर्पणात्।

दौर्बल्यान्मर्मणो घाताद्वन्द्वाच्छुद्ध्यतियोगतः॥2॥

अतीसारज्वरच्छर्दिप्रतिश्यायक्षतक्षयात्।

रक्तपित्तादुदावर्ताद्विसूच्यलसकादपि॥3॥

पाण्डुरोगाद्विषाञ्चैव प्रवर्तते गदाविमौ।

निष्पावमाषपिण्याकतिलतैलनिषेवणात्॥4॥

पिष्टशालूकविष्टम्भिवादिहिरुभोजनात्।

जलजानूपपिशितदध्यामक्षीरसेवनात्॥ 5 ॥

अभिष्यन्नुपचाराच्च श्लेष्मलानां च सेवनात्।

कण्ठोरसः प्रतीघाताद्विबन्धैश्च पृथग्विधैः॥ 6 ॥ Ca.Ci. – 17/11-16

विदाहिरुविष्टम्भिरुक्षाभिष्यन्दिभोजनैः।

शीतपानासनस्थानरजोधूमानिलानलैः॥ 3 ॥

व्यायाम कर्मभाराध्ववेगाघातापतर्पणैः।

आमदोषाभिघातस्त्री क्षयरोगप्रपीडनैः॥ 4 ॥

विषमाशनाध्यनशनैस्तथा संशमनैरपि।

हिक्का श्वासश्च कासश्च नृणां समुपजायते॥ 5 ॥ Su.Ut. – 50/3-5

विदाहिरुविष्टम्भिरुक्षाभिष्यन्दिभोजनैः।

शीतपानाशनस्थानरजोधूमातपानिलैः॥ 1 ॥

व्यायामकर्मभाराध्ववेगाघातापतर्पणैः।

Ma.Ni. – 12/1-2

- ❶ Excess exposure to dust
- ❶ Excess exposure to **smoke**¹
- ❶ Excess exposure to cold air
- ❶ Residing in very cold environment for most of the occasions
- ❶ Excess intake of **ice cold water**²
- ❶ **Excess exercise**³
- ❶ Excess indulgence in **sexual intercourse**⁴
- ❶ Excess intake of **dry food substances**⁵
- ❶ Frequent consumption of food substances at **improper time and in inappropriate quantity**⁶
- ❶ When an individual is suffering **Amapradosha (acute affliction of alimentary canal)**⁷

G
K

- 1 Each nostril is having certain amount of hairs at the entrance to prevent the movement of smoke and dust particles. The second phase of defence is presence of turbinates in the nasal cavity. The third phase of defence is cilia present in the respiratory mucosa and the fourth phase of defence is mucus secreted by the mucous membranes of respiratory system. These are different levels of defence which prevent the entry of dust and smoke particles of various size which are usually less than 6 micrometers of size. Particles which are of smaller than 0.5 micrometers in diameter are usually suspended in alveolar air and are expelled by expiration. Cigarette smoke particles are usually of 0.3 micrometer and 1/3rd of these particles are precipitated in the alveoli and rest of the particles are suspended in the air and expelled out via expired air.
These particles are considered as irritants to the respiratory airways and thus they are responsible for the manifestation of cough and breathlessness.
- 2 When the person is exposed to cold environment in the form of cold air or living in cold room or consumption of cold food substances etc, that will lead to the entry of cold air in the respiratory passages. This will be responsible for the contraction of respiratory mucosa leading to dryness of the respiratory mucosa. The same pathological event occurs when a person is breathing through open mouth in case of nasal obstruction or in case of intubation through tracheostomy will lead to cooling and drying of lower respiratory system leading to serious lung crusting and lung infection.
- 3 It's a question mark, how excess exercise can lead to the occurrence of breathlessness. In the textbook of physiology written by Guyton and Hall, normal oxygen consumption for a young man at rest is 250 ml/ minute. This quantity may increase to 3600 ml/ minute in untrained average male; athletically trained male increases upto 4000 ml/minute and male marathon runner, oxygen consumption may increase to 5100 ml/ minute.
As far as pulmonary ventilation is concerned, at rest pulmonary ventilation is occurring about 9 to 11 litres/ minute and during maximal exercise, the pulmonary ventilation increases about 100 to 110 litres/ minute and maximal breathing capacity increases about 150 to 170 litres/ minute.
When the person is performing exercise with such maximal exercise, the person will also breathe through open mouth. The other determinants of changes in pulmonary ventilation are exercise at higher altitudes and exercise under very hot conditions and pre existing respiratory disease.
 - This will lead to the entry of numerous respiratory irritants and microbes into the respiratory passages lead
- 4 When an individual is having previous history of respiratory disease, then performing the sexual intercourse by the individual having compromise in respiratory function, he will have air hunger, soon after doing sexual intercourse. At this juncture, the person suffering from the respiratory disease will have breathing through open mouth. The person will have pursed lip breathing or deep breathing through open mouth, this will lead to entry of dust, smoke or allergen into the respiratory airways, leading to the exacerbation of respiratory symptoms.
- 5 When a person consumes excess amount of dry food substances, this may irritate vagus nerve, part of parasympathetic nervous system leading to the occurrence of reflex cough. Vagus nerve is considered to be afferent nerve pathway for the respiratory system as well as most of medulla oblongata which in turn leads to activation of both alimentary canal as well as respiratory system leading to the appearance of cough. Breathlessness is observed as an accompaniment of cough.
- 6 Let us take the example of consumption of food substances at large quantity at prior to the digestion of previously ingested meal, this may lead to poor or delayed digestion. This may lead to metabolic acidosis either ketoacidosis or lactic acidosis or both. The further state of this acidosis is respiratory alkalosis. This is characterised by breathing discomfort.
 - Suppose if the person consumes very less amount of food at a particular point of time, it will lead to sequential depletion of carbohydrate, fats and proteins. This may lead to state of metabolic acidosis (ketoacidosis via fat metabolism and lactic acidosis via metabolism in the muscles). Further may develop respiratory alkalosis.
 - If the person takes the food substances too early, it may act as Adhyashana or Atyashana (too much amount of food), this may act like effect produced i.e. prolonged digestion of food, which may lead to metabolic acidosis and later there will be respiratory alkalosis.
 - If the person consumes food after the peak time of appetite, then that may lead to switching over of metabolic acidosis which further may lead to respiratory alkalosis.
- 7 When a person is suffering from Amapradosha (acute affliction of alimentary canal), there will be inflammation of alimentary canal, this

- ❖ When an individual is suffering from distension of abdomen⁸
- ❖ When a person is exposed to dry environment⁹
- ❖ When an individual is undergoing the treatment measures leading to depletion of body tissues¹⁰
- ❖ When an individual is having severe debility¹¹
- ❖ When a person is having injury to Marma (vital points)¹²
- ❖ Dvandva¹³
- ❖ When an individual is undergoing excess Shodhana (purificatory therapy) line of treatment excessively¹⁴
- ❖ When an individual is suffering from Atisara, Jvara, Chardi, Pratishyaya, Kshata, Kshaya, Raktapitta,

Udavarta, Visuchika and Alasaka, Panduroga and Visha¹⁵

- ❖ If a person consumes Nishpava (beans), blackgram, Pinyak (oil cake), substances prepared from sesamum oil, substances prepared from flour, Shaluka (esculent root of different kinds of lotus), substances which hampers the movement of Vata Dosha in the Koshttha (alimentary canal), substances which cause Vidaha (burning sensation in the chest and thirst),¹⁶
- ❖ Substances which are heavy to digest
- ❖ If the person consumes the meat of aquatic and marshy animals
- ❖ If the person consumes curds in excess quantity
- ❖ If the person consumes milk without proper boiling
- ❖ If the person consumes more amount of Abhishyandi

will lead to the either vomiting or diarrhoea. Due to this, there will be rapid loss of sodium or potassium or calcium ions, this will lead to electrolyte imbalances. This may also lead to metabolic acidosis. Metabolic acidosis can result from

1. Failure of kidneys to excrete metabolic acids normally formed in the body;
2. Formation of excess quantities of metabolic acids in the body;
3. Addition of metabolic acids to the body by ingestion and infusion of acids to the body;
4. Loss of base from the body fluids

When the individual is suffering from inflammation of alimentary canal, that will lead to formation of metabolic acids in the body due to anaerobic metabolism. One of the symptoms observed in metabolic acidosis, is increased ventilation rate.

- 8 When an individual is having distension of the abdomen, this may lead to the pressing of the diaphragm towards the heart and lungs to compress. This compression will lead to the compromise in respiratory and cardiac function. One of the earliest clinical manifestation in this situation is breathlessness. In this situation, the ideal line of treatment could be to facilitate the peristaltic movement of the intestine which in turn leads to the decrease in the push of diaphragm towards thorax.
- 9 When an individual is exposed to dry environment, that will lead to decreased secretion in the respiratory passages. In such situation,
- 10 When the person is undergoing the treatment measures which lead to depletion of the body tissues such as Langhana therapy, that leads to the sequential depletion of carbohydrates, fats and proteins, associated with electrolyte imbalance in the form of hyponatremia or hypokalemia or hypocalcemia. This will lead to the formation of ketoacids and lactic acidosis, which is considered as metabolic acidosis. One of the major symptom manifested in metabolic acidosis is hyperventilation with increased ventilation rate.
- 11 When the individual is suffering from severe debilitating illnesses will have extreme nutrient loss from the body such as malignancy or infection such as tuberculosis etc. In this situation, the person will have sequential depletion of carbohydrates, fats and proteins associated with electrolyte imbalances such as hyponatremia or hypokalemia or hypocalcemia. This state is responsible for ketoacidosis and lactic acidosis, a state of metabolic acidosis. One of the cardinal feature of metabolic acidosis is hyperventilation with increased ventilation rate.
- 12 When an individual is having trauma to any of the Marma (vital points), that will lead to extreme fluid and nutrient loss from the body. In this situation, the person will have sequential loss of carbohydrates, fats and proteins, hyponatremia, hypokalemia or hypocalcemia. This will lead to the formation of ketoacidosis or lactic acidosis, a state of metabolic acidosis. One of the main clinical feature of metabolic acidosis is hyperventilation with increased ventilation rate.
- 13 Ayurveda textbooks described the term Dvandva as Sharirika Guna or Vimshati Guna or 20 qualities, with 10 qualities having exactly opposite nature. For example, Guru (heavy to digest) and Laghu (light to digest) qualities; Shita (cold in touch and potency) and Ushna (hot in touch and potency) qualities etc. If an individual is exposed to such type of qualities at a particular time, that will lead to varied functions inside the body. Alimentary canal and respiratory system are the two systems which are directly coming into exposure to these kinds of Dvandva Guna. Let us take the example of exposure of Shita and Ushna to the respiratory system, Shita quality exposure will lead to the reduced secretion of respiratory mucosa whereas proper secretion of mucus may occur on exposure to hot quality exposure to the respiratory system. If the duration of exposure of Shita quality is more than the exposure of Ushna quality to the respiratory passages, the person may show respiratory symptoms.
- 14 When an individual is undergoing the Shodhana (purification) therapy excessively, that will lead to excessive fluid loss from the alimentary canal, which will be responsible for the loss of water from intravascular, interstitial and intracellular compartments sequentially. As the person is having restricted diet, there will be sequential depletion of carbohydrates, fats and proteins, along with electrolyte imbalances such as hyponatremia, hypokalemia or hypocalcemia. This will lead to a state of ketoacidosis or lactic acidosis, a state of metabolic acidosis. One of the major clinical feature of metabolic acidosis is hyperventilation associated with increased ventilation rate.
- 15 Most of the diseases mentioned in this sentence are responsible for profound depletion of nutrients and electrolytes from the body. If above mentioned diseases are observed in the patient for longer time, there will be sequential diminution of carbohydrates, fats and proteins along with hyponatremia or hypokalemia or hypocalcemia. This will lead to a state of ketoacidosis or lactic acidosis, a state of metabolic acidosis. This pathological process is responsible for the manifestation of hyperventilation with increased ventricular rate.
- 16 Excess consumption of these substances will lead to the vitiation of Vata or Kapha Dosha depending upon the nature and amount of consumption of food substances. Whenever there is increase of Vata Dosha, that is responsible for the reduced secretion in the respiratory airways and when there is vitiation of Kapha Dosha, that is responsible for the occlusion of respiratory tract secondary to increased mucus secretion.

- Ahara (substances which cause obstruction in the Srotas)
- If the person consumes more amount of *Kapha* vitiating substances¹⁷
- Suppression of natural urges¹⁸
- If the person consumes excess amount of food before the digestion of previously ingested meals¹⁹.

Samprapti of Shvasa

यदा स्रोतांसि संरुध्य मारुतः कफपूर्वकः।
विष्वजति संरुद्धस्तदा श्वासान्करोति सः॥४५॥
विहाय प्रकृतिं वायुः प्राणोऽथ कफसंयुतः।
श्वासयत्यूर्ध्वगो भूत्वा तं श्वासं परिचक्षते॥४॥

Ca.Ci. – 17/45

Su.Ut. – 51/4

- When an individual is exposed to etiological factors which vitiate *Vata* and *Kapha* Dosha, these Dosha get vitiated.
- These factors will lead to the formation of *Ama* in the body which in turn leads to the reverse i.e. upward movement of *Vata* Dosha leading to the manifestation of *Shvasa*.

Samprapti Ghataka

- Dosha :** *Vata* and *Kapha* Dosha vitiation with secondary involvement of *Pitta*
- Dushya :** *Rasa*

Purvarupa of Shvasa

आनाहः पार्श्वशूलं च पीडनं हृदयस्य च।
प्राणस्य च विलोमत्वं श्वासानां पूर्वलक्षणम्॥२०॥
प्राग्रूपं तस्य हृत्पीडा भक्तद्वेषोऽरतिः परा।
आनाहः पार्श्वयोः शूलं वैरस्यं वदनस्य च॥६॥

Ca.Ci. – 17/20

Su.Ut. – 51/6

- Distension of the abdomen²⁰
- Pricking type of pain on both sides of the chest
- Discomfort in the centre of the chest or heart region²¹
- Reverse movement of *Pranavata*²²
- Tastelessness²³
- Restlessness²⁴
- Abnormal taste perception²⁵

- The consumption of these food substances are responsible for the vitiation of *Kapha* Dosha by default. These substances will lead to the increase of *Snigdha*, *Shita*, *Guru*, *Shlakshna* qualities in the body and this increase in the respiratory system will be responsible for the appearance of breathlessness in the suffering individual.
- Let us take the example of urge of defaecation. When an individual is having the urge of defaecation and he suppresses the urge will lead to the vitiation of *Apanavata* in *Pakvashaya* and this will lead to the reverse of *Apanavata* and this mechanism is termed as *Udavarta*. In due course of time, upward movement of *Apanavata* afflicts *Samanavata* and *Pranavata* sequentially. Thus, breathlessness can be observed as a late feature in suppression of defaecation.
- The term *Adhyashana* refers to ingestion of meals before the digestion of previously ingested meal. As per fundamentals of *Ayurveda*, there will be vitiation of *Kapha* Dosha along with dominance of *Vata* Dosha. When an individual is consuming excess intake of food before the digestion, that will lead to the accumulation of partially digested foods in the storage. This will lead to reverse peristaltic movement leading to aspiration pneumonia if gastric contents enter the respiratory passages, thus leading to the mechanisms of cough and breathlessness.
- When the individual is presented with distension of the abdomen, it indicates poor peristaltic movements. When there is reduced peristaltic movements, it implies the person is having activation of parasympathetic nervous system. This activation will be responsible for generalised action of parasympathetic nervous system. In the alimentary canal, this will lead to increased alimentary canal secretions. In the respiratory system, the activation of parasympathetic nervous system will lead to the increased mucus secretion in the upper and lower respiratory tract. Thus, accumulation of mucus in the respiratory tract may become a triggering factor for breathlessness.
- Two types of chest pain are of significance – centralised and lateralised chest pain. Centralised chest pain where the person feels pain in the retrosternal region is observed in the patients who have mediastinal involvement and lateralised chest pain where the person may complain of pain in one or both sides of the chest is observed when pleural involvement is observed in respiratory disease.
- Normal *Pranavata* has the functions such as *Nishvasa* (inspiration), *Anna Pravesha* (deglutition) as far as movement of *Pranavata* is concerned. When there is reverse movement of *Pranavata*, the person may have the difficulty in breathing and the person may have regurgitation, nausea or vomiting.
- When an individual is presented with breathlessness, it will be associated with autonomic dysfunction. If the person is having activation of sympathetic nervous system, there will be dryness of the mouth, throat and respiratory passages. When there is dryness of the mouth, tastelessness is evident, as there is no salivary secretion.
- This is attributable to the electrolyte imbalance. This is usually observed in case of extreme emotional lability such as anxiety, grief, sorrow or anger etc. When the individual has emotional disturbances, he will breathe through open mouth and this may lead to electrolyte imbalance. This can be attributable to hyperventilation syndrome. Hyperventilation syndrome is seen in younger women and it should be differentiated from panic disorder.
- When an individual is presented with breathlessness, it will be associated with autonomic dysfunction. If the person is having activation of

Types of Shvasa

5 types of Shvasa as per Charaka, Sushruta and Madhavakara

1. Mahashvasa
2. Urdhva Shvasa
3. Chinna Shvasa
4. Tamaka Shvasa
5. Kshudra Shvasa

Dosha dominance of Shvasa as per Madhavakara

(वाताधिको भवेत् क्षुद्रस्तमकस्तु कफोद्धवः।

कफवाताधिकश्चैव संसृष्टश्छिन्नसंज्ञकः।

श्वासो मारुतसंसृष्टो महानूर्ध्वस्ततो मतः॥1॥) Ma.Ni. – 12/ 15-1

- Kshudra Shvasa is having Vata dominance; Tamaka Shvasa is having Kapha dominance; Chinna Shvasa is having Kapha Vata dominance; Maha Shvasa and Urdhva Shvasa is having Vata dominance.

Mahashvasa²⁶

उद्ध्वयमानवातो यः शब्दबहुः खितो नरः।

उच्चैः श्वसिति संरुद्धो मत्तर्षभ इवानिशम्॥4६॥

प्रनष्टज्ञानविज्ञानस्तथा विभ्रान्तलोचनः।

विकृताक्ष्याननो बद्धमूत्रवर्चा विशीर्णवाक्॥4७॥

दीनः प्रश्वसितं चास्य दूराद्विज्ञायते भृशम्।

महाश्वासोपसृष्टः स क्षिप्रमेव विपद्यते॥4८॥

Ca.Ci. – 17/46-48; Ma.Ni. – 12/18-20

विसंज्ञः पार्श्वशूलार्तः शुष्ककण्ठोऽतिघोषवान्।

संरुद्धनेत्रस्वायम्य यः श्वस्यात् स महान् स्मृतः॥ Su.Ut. – 51/12

- In this type of Shvasa, Vata Dosha is vitiated so that breathing discomfort produced during this stage, there will be loud sound during breathing. The sound can be heard from far away distance like the sound produced by intoxicated bull.
- There will loss of orientation in relation to place and time
- Restlessness
- Rapid movement of eyes
- Patient will be having abnormal movement of the eyes and mouth
- Constipation and retention of urine
- The person appears sick or pathetic and takes lots of stress for breathing or struggles for breathing.

- This type of Shvasa is very difficult to cure and it is a fatal sign and the patient dies soon.

As per Sushruta,

- Altered sensorium
- Pain in sides of the chest
- Dryness of the throat
- Loud sounds during respiration
- Swelling in the eyes

Urdhva Shvasa²⁷

दीर्घं श्वसिति यस्तूर्ध्वं न च प्रत्याहरत्यधः।

श्लेष्मावृतमुखस्रोताः क्रुद्धगन्धवहार्दितः॥4९॥

ऊर्ध्वदृष्टिर्विशयंश्च विभ्रान्ताक्ष इतस्ततः।

प्रमुह्यन् वेदनार्तश्च शुष्कास्योऽरतिपीडितः॥5०॥

ऊर्ध्वश्वासे प्रकुपिते ह्यधःश्वासो निरुध्यते।

मुह्यतस्ताम्यतश्चोर्ध्वश्वासस्तस्यैव हन्त्यसून्॥5१॥

Ca.Ci. – 17/49-51; Ma.Ni. – 12/21-23

मर्मस्वायम्यमानेषु श्वसम्मूढो मुहुश्च यः।

ऊर्ध्वप्रेक्षी हतरवस्तमूर्ध्वश्वासमादिशेत्॥1३॥

Su.Ut. – 52/13

- The person will be having long and deep inspiration, but expiration is difficult due to accumulation of Kapha in the Pranavaha Srotas.
- The patient will have upward gaze
- Eyes move aimlessly in all directions
- Altered state of consciousness
- Pain in the chest
- Dryness of mouth
- Restlessness
- Confusion
- Darkness in front of the eyes
- The patient may lose his life shortly

As per Sushruta,

- Affliction of Hridaya, Basti and Shiras etc Marma can be observed in this type of Shvasa.
- Breathing discomfort
- Altered state of consciousness frequently
- Upward gaze
- Loss of sound from the throat

Chinna Shvasa²⁸

यस्तु श्वसिति विच्छिन्नं सर्वप्राणेन पीडितः।

parasympathetic nervous system, that will lead to hypersecretion of saliva. This may lead to abnormal taste perception.

26 As in case of Atisara, Chardi, Visuchika and Shodhana Atiyoga, affliction of alimentary canal is seen. This will be associated with extreme fluid loss and electrolyte imbalance as well. During the progression of this disease, Dourbalya, Sharira Rukshata and Ati Apararpana can be observed which results from undernourishment, hyponatremia in particular.

27 Hridaya Marmabhighata, Pandu Roga and Raktapitta spectrum of diseases leading to cardiac failure where in pulmonary oedema may be the result due to backward failure..

28 Shiro Marmabhighata (injury to posterior cranial fossa), Visha Sevana (exposure to toxic substances) where in respiratory centre affliction

न वा श्वसिति दुःखार्तो मर्मच्छेदरुगर्दितः॥52॥

आनाहर्षेदमूर्च्छात दह्यमानेन वस्तिना।

विताक्षः परिक्षीणः श्वसन् रक्तकलोचनः॥53॥

विचेताः परिशुष्कास्यो विवर्णः प्रलपन्नरः।

छिन्नश्वासेन विच्छिन्नः स शीघ्रं प्रजहात्यसूत 54

Ca.Ci. - 17/52-55; Ma.Ni. - 12/24-26

आध्मातो दह्यमानेन वस्तिना सरुज नरः।

सर्वप्राणेन विच्छिन्नं श्वस्याच्छिन्नं तमादिशेत्॥11॥ Su.Ut. - 51/11

• The word *Chinna* refers to cut or break in the pattern of breathing. That means the person will be having the break or discontinuity in the pattern of breathing. In other words, in this type of *Shwasa*, the person will have rapid phase of breathing sometimes and sometimes he will have slow phase of breathing.

• This pattern indicates the involvement or injury of *Marma*.

• Distension of the abdomen

• Excess sweating

• Altered state of consciousness

• Burning sensation in lower abdomen

• Instability in the movement of the eyes

• Depletion of the body tissues

• Breathing discomfort

• Reddish discolouration of only one eye

• Loss of consciousness

• Dryness of mouth

• Discolouration of the body parts

• Irrelevant and incomprehensible speech

• The patient will lose his life soon if no proper care is being given.

As per Sushruta,

• Distension of the abdomen

• Burning sensation in the lower abdomen

• Pain in the lower abdomen

• Discontinuous breathing pattern

Tamaka Shwasa²⁹

प्रतिलोमं यदा वायुः स्त्रोतांसि प्रतिपद्यते।

ग्रीवां शिरश्च संगृह्य श्लेष्माणं समुदीर्य च॥55॥

करोति पीनसं तेन रुद्धो घुघुरुकं तथा।

अतीव तीव्रवेगं च श्वासं प्राणप्रपीडकम्॥56॥

प्रताम्यत्यतिवेगाच्च कासते सन्निरुध्यते।

प्रमोहं कासमानश्च स गच्छति मुहुर्मुहुः॥57॥

श्लेष्मण्यमुच्यमाने तु भृशं भवति दुःखितः।

can be seen where abnormal and non uniform breathing patterns are observed.

29 Kasa, Pratishyaya, Kshatakshaya (injury to chest), Manasa Dosha (unstable and weak mental functions) leading to the abnormal functioning of the respiratory system. Here, the causes of respiratory system dysfunction comes into play in the manifestation of the disease.

तस्यैव च विमोक्षान्ते मुहूर्तं लभते सुखम्॥58॥

अथास्योसते कण्ठः कृच्छ्राच्छ्वनोति भाषितुम्।

न चापि निद्रां लभते शयानः श्वासपीडितः॥59॥

पार्श्वे तस्यावगृह्णाति शयानस्य समीरणः।

आसीनो लभते सौख्यमुष्णं चैवाभिनन्दति॥60॥

उच्छ्रिताक्षो ललाटेन विद्यता भृशमर्तिमान्।

विशुष्कास्यो मुहुः श्वासो मुहुश्चैवावधयते॥61॥

मेघाम्बुशीत प्राग्वातैः श्लेष्मलैश्चाभिवर्धते।

स याप्यस्तमकश्वासः साध्यो वा स्यान्नवोत्थितः॥62॥

Ca.Ci. - 17/55-62; Ma.Ni. - 12/27-34

तृप्त्रस्वेदवमथुप्रायः कण्ठघुघुरिकान्वितः।

विशेषादुर्दिने ताम्येच्छासः स तमको मतः॥8॥

घोषेण महताऽऽविष्टः सकासः सकफो नरः।

यः श्वसित्यवलोल्लङ्घि सुप्तैस्तमकपीडितः॥9॥

स शाम्यति कफे हीने स्वपतश्च विवर्धते।

मूर्च्छाज्वराभिभूतस्य ज्ञेयः प्रतमकस्तु सः॥10॥ Su.Ut. - 51/8-10

• The word *Tamaka* refers to darkness. In *Tamaka Shwasa*, the person will be having darkness in front of the eyes associated with breathlessness.

• The vitiated *Vata* will move upwards and occludes the *Srotas*, causing stiffness of the nape of neck and head and provoking *Kapha Dosha*.

• Running nose

• Creptitation or musical sound is observed due to constriction in the passage of air in the *Pranavaha Srotas*

• There will be breathing discomfort which is associated with severe distress and pain.

• There will be severe bouts of cough with little amount of sputum will be spitted out associated with darkness in front of the eyes. When the little amount of sputum is expelled, then the person will have relief for a shorter time period, again the person will have cough and breathing discomfort and these episodes will continue.

• There will be hoarseness of voice due to frequent coughing.

• Speaking will be difficult due to breathing discomfort.

• Attaining recumbent posture to get sleep will be difficult and the person has to take sitting posture during the episode of breathlessness.

• The patient always prefers to have warm or hot food substances and drinks.

• The patient will have swelling around eyes.

• Profuse sweating over forehead.

• Dryness of mouth

• Frequent episodes of breathlessness

- Worsening of the symptoms of *Shvasa* is seen on exposure to cloudy days, rainy season, winter season, exposure to cold air and on consumption of *Kapha* vitiating etiological factors.

As per Sushruta,

- Thirst
- Excess sweating
- Vomiting
- Typical sound from the throat and loud sound
- Worsens on abnormal days
- Cough with sputum
- Aversion to take food
- When sputum is expelled, the person gets relief from breathing discomfort
- On lying down posture, there will be worsening of symptoms.

ज्वरमूर्च्छापरीतस्य विद्यात् प्रतमकं तु तम्।
उदावर्तजोऽजीर्णक्लिन्नकायनिरोधजः॥६३॥
तमसा वर्धतेऽत्यर्थं शीतैश्चाशु प्रशाम्यति।
मज्जतस्तमसी वाऽस्य विद्यात् संतमकं तु तम्॥६४॥

Ca.Ci. – 17/63, 64

Types of Tamaka Shvasa

2 types as per *Charaka* depending upon the stage of the illness.

- Pratamaka Shvasa** – associated with fever and altered state of consciousness
- Santamaka Shvasa** – worsens when the person is having distension of abdomen, exposure to dust, in a state of indigestion, in elderly people, suppression of urges, and in darkness or night time. The administration of cold measures will improve the situation in *Santamaka Shvasa*.

Kshudra Shvasa³⁰

रूक्षयासोद्भवः कोष्ठे क्षुद्रो वात उदीरयन्।
क्षुद्रश्वासो न सोऽत्यर्थं दुःखेनाङ्गप्रबाधकः॥६५॥
हिनस्ति न स गात्राणि न च दुःखो यथेतरे।
न च भोजनपानानां निरुणच्युचितां गतिम्॥६६॥
नेन्द्रियाणां व्यथां नापि कांचिदापादयेद्भुजम्।
स साध्य उक्तो बलिनः सर्वे चाव्यक्तलक्षणाः॥६७॥

Ca.Ci. – 17/65-67

- The word *Kshudra* refers to mild or minor or least important. Thus, when we see the other types of *Shvasa*, this type of *Shvasa* is having less severe symptoms and patient will recover soon.
- This occurs when an individual consumes dry substances and performing slight physical exertion. This will lead to the vitiation of *Vata* in the *Koshtha* (alimentary canal) to move upwards and leads to occurrence of breathing discomfort.
- Slight amount of rest leads to recovery from the discomfort. No treatment is necessary for this type of *Shvasa*.
- The person will have no difficulty in eating and drinking.
- This disease may not stay longer as there is naturally vitiated *Dosha* in the body and they will get pacified easily and because of mild symptoms.

Sadhyasadhya³¹

क्षुद्रः साध्यो मतस्तेषां तमकः कृच्छ्र उच्यते।

त्रयः श्वासा न सिध्यन्ति तमको दुर्बलस्य च॥४०॥ Ma.Ni. – 12/40

- Kshudra Shvasa* is *Sadhya* (easily curable)
- Tamaka Shvasa* is *Krichra Sadhya* (curable with difficulty)
- Maha Shvasa*, *Urdhva Shvasa* and *Chinna Shvasa* are *Asadhya* (incurable)

Differential diagnosis of Shvasa

Previous history of the illness:

- *Atisara* (diarrhoea), *Chardi* (vomiting), *Visuchika* (gastro enteritis), *Shuddhati yoga* (excess purification procedure) are the conditions where in involvement of alimentary canal are seen in the form of excessive vomiting or *diarrhoea*. Excess fluid loss along with electrolyte imbalance will eventuate during these pathological conditions. *Dourbalya* (whole body weakness), *Sharira Rukshata* (dryness of the body parts or loss of skin turgor) and *Ati Apararpana* (excess emaciation) are the conditions leading to malnutrition or undernourishment. These are the diseases in which metabolic disturbances may occur during the course of the illness; loss of alkali in *diarrhoea*, malnutrition and starvation leading to ketosis in protracted vomiting leading to metabolic

³⁰ Ampradosha, Anaha, Udavarta, Alasaka where in distension of the abdomen is seen. This is indicative of pushing of the diaphragm towards the chest region leading to breathlessness.

³¹ This is determined by the duration of the illness, nature of the symptoms, severity of the symptoms. *Kshudra Shvasa* is considered to be milder form among the types of *Shvasa*. That is the reason, this type of *Shvasa* is easily curable. *Tamaka Shvasa* is an episodic illness, depending upon the duration of the illness, exposure to the degree of the aetiological factor, this type of *Shvasa* is curable with difficulty. *Maha Shvasa*, *Urdhva Shvasa* and *Chinna Shvasa* are considered as incurable as the duration of the illness is longer and persistent and symptoms will be persisting throughout the duration of the illness. That is why, the first three varieties of *Shvasa* mentioned are considered to be incurable

disturbances, hyponatraemia in particular. It can be considered that these disease entities predispose *Mahaswasa*.

- *Hridaya Marmabhighata* (myocardial infarction), *Pandu roga* (anemia) and *Raktapitta* (bleeding diathesis) group of diseases lead to cardiac failure resulting in pulmonary oedema. These diseases may predispose *Urdhwa Swasa*.

- *Siro Marmabhighata* (head injury, injury to the posterior cranial fossa in particular), *Visha Sevana* (exposure to toxic substances) are the conditions where dysfunction of respiratory centre is noticed which is responsible for the abnormal breathing pattern. These may predispose *Chinna Swasa*.

- *Kasa* (cough), *Pratishyaya* (rhinitis), *Kshatakshaya* (injury to the chest) and *Manasa Dosha* (unstable or weak mental functions) are resulting in the abnormal function of respiratory system. These may predispose to *Tamaka Swasa*. Definitely the same factors may initiate the obstructive airway disease i.e. bronchial asthma.

- *Amapradosha* (Diseases of the alimentary canal due to improper digestion), *Anaha* (distension of the abdomen), *Udavarta* (Abnormal reversal in the course of *vata* within the alimentary tract), *Alasaka* (distension of the abdomen due to the accumulation of undigested food in the *Amashaya* i.e. stomach) are responsible for the distension of the abdomen. The above mentioned conditions are primarily responsible for impaired movement of diaphragm. As it is primary muscle of respiration breathlessness will be evident. These features are observed in the patients suffering from *Kshudraswasa*.

- **Duration of the illness:** Duration of the illness is another criterion by which differentiation of types of *Swasaroga* can be simplified. Short duration of the illness is the typical feature of *Mahaswasa*, *Urdhwaswasa*, *Chinna Swasa*. In contrast to this, patient of *Tamaka Swasa* or obstructive airway disease usually gives a long history of illness. The duration of *Kshudraswasa* or respiratory distress due to abdominal conditions is variable.

- **Mode of onset:** Like the duration of the illness, mode of onset of *Swasa* is also useful in the differential diagnosis of *Swasa*. *Shighra Utpatti* or sudden onset of *Swasa* or breathless is suggestive of *Mahaswasa*, *Urdhwaswasa* as well as *Chinna Swasa*. *Chira Utpatti* or gradual onset of breathlessness is most commonly observed in *Tamaka Swasa* and *Kshudra Swasa*. But sudden attack of *Swasa* can also be frequently observed in the patients of *Tamaka Swasa*.

- **Character of Shvasa: Character of Swasa:** This is one of the most important and useful criteria to differentiate the *Swasa roga*.

- **Prana Vilomata:** In *Tamaka Swasa* and *Kshudra Swasa* there will be laboured breathing with the predominance of expiration. Due to bronchospasm, patient has to put more effort during expiration in bronchial asthma. Inspiration is restricted due to reduced diaphragmatic movement in case of abdominal distension and hence expiration is predominant.

- **Deergham Ati Ghoshavan Swasiti - Sudirgha Sashabda Swasa** - i.e. loud and prolonged respiration that can be heard from a distance is characteristic of *Mahaswasa*. Similarly in case of metabolic acidosis, hyperventilation is evident; breathing is intense with hissing and known as Kussmaul's breathing.

- **Sudirgham Urdhwa Swasa Nishwasa Rahita - Hraswa Nishwasa yukta** - in this breathing abnormality, patient shows prolonged expiration with no or shallow inspiration and is diagnostic of *Urdhwa Swasa*. Also in acute pulmonary oedema there will be asphyxiation as the alveoli that are flooded with fluid and breathing is rapid and no transfer of oxygen is possible.

- **Nishwasya Punah Kshanantena Shwasiti - Vicchinnam Antara Antara Shwasiti** - Phase of respiration and no respiration occurring alternatively is pathognomonic of *Chinna Swasa*. In comparison to this when the disturbed functioning of respiratory centre develops, the respiration will be irregular. Alternative gradual appearance of phase of *apnea* and hyperventilation is characteristic of Cheyne Stokes breathing. Waxing and waning of respiration is also suggestive of this periodic breathing.

- **Course of the illness:** Course of the illness is not uniform in different types of *Swasa roga*. The course of the illness may be *Nirantara* (persistent) or *Santara* (associated with *Vega*). *Nirantara* (persistent) symptoms seen in *Mahaswasa*, *Urdhwaswasa* and *Chinnaswasa* which may result in the death of the patient. In case of metabolic disturbances, acute pulmonary oedema and disturbance of respiratory centre, the symptoms are persistent. Definitely the prognosis is poor if energetic treatment is not being adopted. Occasionally in *Tamaka Swasa*, the course of the illness may be prolonged with continuous manifestation of symptoms usually associated with exacerbation and remission. *Santara Vegavan* or episodic symptoms and long course of the illness are typically suggestive of *Tamaka Swasa*. The course of the illness is not specific in *Kshudra Swasa*. It may be continuous or intermittent depending upon the inducing or triggering factor.

- **Severity of illness:** Severity may be *Mridu*, *Daruna* or *Aniyamita* in different types of *Swasaroga*. The

in the patients who are having disturbance in respiratory centre and are depending upon the causative illness.

- The symptoms like *Sakapha Kasa* (productive cough), *Pinasa* (running nose), *Gurghuraka* (rattling), *Kanthodhuwamsa* (soreness of throat), *Vishushkasya* (dryness of mouth), *Lalatasweda* (sweating in the forehead region), *Ut shunaksha* (oedema around eyes) are seen in the patient of *Tamaka Swasa*. All these symptoms are observed in the patient of bronchial asthma.

- Associated symptoms are usually absent in *Kshudraswasa*.

- Modifying factors:** Severity of *Swasa* is frequently altered by some aggravating and relieving factors. By identifying these factors one can differentiate the types of *Swasa*. Aggravation of the *Swasa* occurs in case of *Amuchyamana Shleshma* (lack of expectoration), *Shayane Sthiti* (patient in recumbent posture), *Meghambu Shita Pragvata* (exposure to cold and cloudy weather), *Shleshmala Ahara Sevana* (consumption of food and drinks which increase the *Kapha Dosha* in the body), *Raja Sevana* (exposure to dust), *Dhuma Sevana* (exposure to smoke), *Ratrisamaya* (night hours), *Manodosha* (emotional factors). Relief from *Swasa* occurs in case of *Shleshma Nishtivana* (expectoration of sputum), *Ushna Upachara* (warm environment and food) and *Asine Sukhanubhava* (more comfort in upright posture).

- **Seasonal variations:** The incurable types of *Swasa* that include *Mahaswasa*, *Urdhwaswasa* and *Chinnaswasa* have fatal outcome, manifests comparatively for a short duration. Hence, seasonal changes do not influence the severity of the illness. In contrast to this, in case of *Tamaka Swasa*, worsening of the symptoms is frequently noted in rainy and winter seasons. In case of *Kshudra Swasa* also, no variation in the severity of the symptoms was noted during the changes in the season. Thus, seasonal influence in the symptoms of *Swasa* is always confirmatory of *Tamaka Swasa*.

श्वासत्वं वेगवद्ध्ववातत्वं:

Madhu, Ma.Ni. – 12/15-1

“श्वासस्तु भस्त्रिकाध्मानसमवातोर्ध्वगामिता”

Madhu, Ma.Ni. – 12/15-1

- As far as the Nidana of Shvasa are concerned, the factors which exacerbate the Vata and Kapha Dosha are responsible for the manifestation of Shvasa. This is a Pranavaha Sroto Vikara afflicting Rasa Dhatu with Sanga and Vimargagamana as the type of Srotodushti involved during the pathogenesis of the disease. This is Amashayasamuttha Vikara.
- Anaha (distension of the abdomen), Parshvashula (pain on sides of the chest), Hridayasya Pidana (discomfort in the chest), Pranasya Vilomata (breathlessness during expiration) are Purvarupa of the disease as per Charaka.
- Upward movement of vitiated Vata is the main cardinal feature of the disease.
- 5 types of Shvasa are mentioned: Mahashvasa, Urdhva Shvasa, Chinna Shvasa, Tamaka Shvasa and Kshudra Shvasa. Among the 5 types, Mahashvasa, Urdhva Shvasa and Chinna Shvasa are incurable; Tamaka Shvasa is treatable with much difficulty as it is considered to be episodic illness; Kshudra Shvasa is easily curable.

Self-Assessment: Questions for Review

Long essay questions

- Explain the Nidana, Samprapti, Purvarupa, Bheda, Lakshana and Sadhyasadyata of Shvasa
- Describe the causes, Samprapti, Lakshana and Bheda of Tamaka Shvasa

Short essay questions

- Write in brief about Maha Shvasa
- Describe Chinna Shvasa
- Write Nidana, Lakshana and Sadhyasadyata of Urdhva Shvasa
- Describe the Lakshana of Tamaka Shvasa
- Describe Dosha Pradhanya observed in different types of Shvasa

MCQs

1. The type of Shvasa which is easily curable?

- a. Kshudra Shvasa;
- b. Maha Shvasa;
- c. Chinna Shvasa;
- d. Urdhva Shvasa

Answer: (a)

2. Which of the following type of Shvasa is not incurable?

- a. Maha Shvasa;
- b. Tamaka Shvasa;
- c. Chinna Shvasa;
- d. Urdhva Shvasa

Answer: (b)

3. Gurghuraka is the specific Lakshana which is observed in which of the following disease?

- a. Tamaka Shvasa;
- b. Chinna Shvasa;
- c. Maha Shvasa;
- d. Urdhva Shvasa

Answer: (a)

4. Santamaka and Pratamaka are the types of

- a. Chinna Shvasa;
- b. Maha Shvasa;
- c. Tamaka Shvasa;
- d. Urdhva Shvasa

Answer: (c)

5. Dosha dominance observed in Tamaka Shvasa are

- a. Vata and Kapha;
- b. Vata and Rajas;
- c. Kapha and Tama;
- d. Pitta and Kapha

Answer: (a)

6. In which type of Shvasa, the shape of the patient's chest becomes like that of chest of Kapota?

- a. Urdhva Shvasa;
- b. Maha Shvasa;
- c. Tamaka Shvasa;
- d. Chinna Shvasa

Answer: (c)

7. Which of the following Lakshana is not observed in Tamaka Shvasa?

- a. Gurghuraka;
- b. Asino Labhate Saukhyam;
- c. Vicchinna Shvasa;
- d. Pinasa

Answer: (c)

8. Which of the following type of Shvasa is considered as Yappa (manageable)?

- a. Maha Shvasa;
- b. Chinna Shvasa;
- c. Urdhva Shvasa;
- d. Tamaka Shvasa

Answer: (d)

9. Dirgha Uchhvasa and Pratihata Nishvasa are seen in

- a. Maha Shvasa;
- b. Urdhva Shvasa;
- c. Chinna Shvasa;
- d. Kshudra Shvasa

Answer: (b)

10. Pratamaka Shvasa is

- a. Tamaka Shvasa with Udavarta;
- b. Tamaka Shvasa with Jvara and Murcha;
- c. Tamaka Shvasa with Ajirna;
- d. Tamaka Shvasa with Pinasa

Answer: (b)

11. As per Charaka, the site of disease Tamaka Shvasa is

- a. Phupphusa;
- b. Amashaya;
- c. Uras;
- d. Nasa

Answer: (b)

12. The nature of Srotodushti observed in Tamaka Shvasa is/are

- a. Sanga only;
- b. Sanga and Vimarga Gamana only;
- c. Vimarga Gamana only;
- d. Atipravritti only

Answer: (b)

13. Vishirna Vak and Vibhranta Lochana are the Lakshana observed in

- a. Chinna Shvasa;
- b. Tamaka Shvasa;

- c. Urdhva Shvasa;
- d. Maha Shvasa

Answer: (d)

14. As per Charaka, which of the following is mentioned along with Shvasa as killer disease which has similar Nidana among both of the diseases?

- a. Kasa; b. Rajyakshma;
- c. Shosha; d. Hikka

Answer: (d)

15. Which of the following is mentioned as Purvarupa of Shvasa as per Charaka Samhita?

- a. Ura Shula; b. Parshva Shula;
- c. Udara Shula; d. Basti Shula

Answer: (b)

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